

ALKERAN TABLETS

Melphalan

QUALITATIVE AND QUANTITATIVE COMPOSITION

Each tablet contains 2 mg melphalan.

PHARMACEUTICAL FORM

Film-coated tablets.

CLINICAL PARTICULARS

Therapeutic Indications

Melphalan tablets are indicated in the treatment of:

- multiple myeloma;
- advanced ovarian adenocarcinoma;

Melphalan tablets may be used in the treatment of:

breast carcinoma: melphalan either alone or in combination with other drugs has a significant therapeutic effect in a proportion of patients suffering from advanced breast carcinoma;

polycythaemia rubra vera: melphalan is effective in the treatment of a proportion of patients suffering from polycythaemia vera.

Posology and Method of Administration

General

Melphalan is a cytotoxic drug which falls into the general class of alkylating agents. It should be prescribed only by physicians experienced in the management of malignant disease with such agents.

Since melphalan is myelosuppressive, frequent blood counts are essential during therapy and the dosage should be delayed or adjusted if necessary (*see Special Warnings and Precautions for Use*).

The absorption of melphalan after oral administration is variable. Dosage may need to be cautiously increased until myelosuppression is seen, in order to ensure that potentially therapeutic levels have been reached.

Thromboembolic events

Melphalan in combination with lenalidomide and prednisone or in combination with thalidomide and prednisone or dexamethasone is associated with an increased risk of venous thromboembolism. Thromboprophylaxis should be administered for at least the first 5 months of treatment especially in patients with additional thrombotic risk factors. The decision to take

antithrombotic prophylactic measures should be made after careful assessment of an individual patient's underlying risk factors (see *Special Warnings and Precautions for Use* and *Undesirable Effects*).

If the patient experiences any thromboembolic events, treatment must be discontinued and standard anticoagulation therapy started. Once the patient has been stabilised on the anticoagulation treatment and any complications of the thromboembolic event have been managed, melphalan in combination with lenalidomide and prednisone or thalidomide and prednisone or dexamethasone may be restarted at the original dose dependent upon a benefit-risk assessment. The patient should continue anticoagulation therapy during the course of melphalan treatment.

Multiple myeloma

A typical oral dosage schedule is 0.15 mg/kg bodyweight/day in divided doses for 4 days repeated at intervals of 6 weeks. Numerous regimens have, however, been used and the scientific literature should be consulted for details.

The administration of oral melphalan and prednisone may be more effective than melphalan alone. The combination is usually given on an intermittent basis.

Prolonging treatment beyond one year in responders does not appear to improve results.

Advanced ovarian adenocarcinoma

A typical regimen is 0.2 mg/kg bodyweight/day orally for 5 days. This is repeated every 4 to 8 weeks, or as soon as the peripheral blood count has recovered.

Carcinoma of the breast

Melphalan has been given orally at a dose of 0.15 mg/kg bodyweight or 6 mg/m² body surface area/day for 5 days and repeated every 6 weeks. The dose was decreased if bone marrow toxicity was observed.

Polycythaemia rubra vera

For remission induction, doses of 6 to 10 mg daily for 5 to 7 days have been used, after which 2 to 4 mg daily were given until satisfactory disease control was achieved.

A dose of 2 to 6 mg once per week has been used for maintenance therapy.

In view of the possibility of severe myelosuppression if melphalan is given on a continuous basis, it is essential that frequent blood counts are taken throughout therapy, with dosage adjustment or breaks in treatment, as appropriate, to maintain careful haematological control.

Paediatric population

Melphalan, within the conventional dosage range, is only rarely indicated in children and absolute dosage guidelines cannot be provided.

Elderly

Although melphalan is frequently used at conventional dosage in the elderly, there is no specific information available relating to its administration to this patient sub-group.

Dosage in renal impairment

Melphalan clearance, though variable, is decreased in renal impairment (*See Special Warnings and Precautions for Use*).

Currently available pharmacokinetic data do not justify an absolute recommendation on dosage reduction when administering melphalan tablets to patients with renal impairment, but it may be prudent to use a reduced dosage initially until tolerance is established.

Contraindications

- Hypersensitivity to the active substance or to any of the excipients
- Lactation

Special Warnings and Precautions for Use

Melphalan IS AN ACTIVE CYTOTOXIC AGENT FOR USE UNDER THE DIRECTION OF PHYSICIANS EXPERIENCED IN THE ADMINISTRATION OF SUCH AGENTS.

Immunisation using a live organism vaccine has the potential to cause infection in immunocompromised hosts. Therefore, immunisations with live organism vaccines are not recommended.

Safe handling of melphalan (*See Safe Handling in Special Precautions for Disposal and Other Handling*).

Monitoring

Since melphalan is a potent myelosuppressive agent, it is essential that careful attention should be paid to the monitoring of blood counts to avoid the possibility of excessive myelosuppression and the risk of irreversible bone marrow aplasia.

Blood counts may continue to fall after treatment is stopped, so at the first sign of an abnormally large fall in leucocyte or platelet counts, treatment should be temporarily interrupted.

Melphalan should be used with caution in patients who have undergone recent radiotherapy or chemotherapy in view of increased bone marrow toxicity.

Renal impairment

Melphalan clearance may be reduced in patients with renal impairment, who may also have uraemic bone marrow suppression. Dosage reduction may therefore be necessary (see *Posology and Method of Administration*), and these patients should be closely observed.

The use of high dose melphalan has the potential to cause acute kidney injury in patients, especially those with underlying renal impairment and potential risk factors for reduced renal function (e.g., concomitant use of nephrotoxic medications, amyloidosis etc).

Mutagenicity

Chromosome aberrations have been observed in patients being treated with the drug.

Carcinogenicity (Second primary malignancy)

Acute myeloid leukaemia (AML) and myelodysplastic syndromes (MDS)

Melphalan, in common with other alkylating agents, has been reported to be leukaemogenic, especially in elderly patients after long combination therapy and radiotherapy. There have been reports of acute leukaemia occurring after melphalan treatment for diseases such as amyloidosis, malignant melanoma, multiple myeloma, macroglobulinaemia, cold agglutinin syndrome and ovarian cancer.

A comparison of patients with ovarian cancer who received alkylating agents with those who did not show that the use of alkylating agents, including melphalan, significantly increased the incidence of acute leukaemia.

Before the start of the treatment, the leukaemogenic risk (AML and MDS) must be balanced against the potential therapeutic benefit, especially if the use of melphalan in combination of thalidomide or lenalidomide and prednisone is considered, as it has been shown that these combinations increase the leukaemogenic risk. Before, during and after treatment doctors must therefore examine the patient by usual measurements to ensure early detection of cancer and initiate treatment if necessary.

Solid tumours

Use of alkylating agents has been linked with the development of second primary malignancy (SPM). In particular, melphalan in combination with lenalidomide and prednisone and, to a lesser extent, thalidomide and prednisone has been associated with the increased risk of solid SPM in elderly newly diagnosed multiple myeloma patients.

Patient characteristics (eg. age, ethnicity), primary indication and treatment modalities (e.g. radiation therapy, transplantation), as well as environmental risk factors (e.g. tobacco use) should be evaluated prior to melphalan administration.

Contraception

Due to an increased risk of venous thromboembolism in patients undergoing treatment with melphalan in combination with lenalidomide and prednisone or in combination with thalidomide and prednisone or dexamethasone, combined oral contraceptive pills are not recommended. If a patient is currently using combined oral contraception, she should switch to another reliable contraceptive method (i.e. barrier method etc). The risk of venous thromboembolism continues for 4-6 weeks after discontinuing combined oral contraception.

It is recommended that men who are receiving treatment with melphalan not father a child during treatment and up to 3 months afterwards and that they have a consultation on sperm preservation before treatment due to the possibility of irreversible infertility as a result of melphalan treatment. The recommended duration of contraception in females should be during treatment and for a period of 6 months following the cessation of treatment (*see Fertility, Pregnancy and Lactation*).

Interaction with Other Medicinal Products and Other Forms of Interactions

Live organism vaccines

Vaccinations with live organism vaccines are not recommended in immunocompromised individuals (*see Special Warnings and Precautions for Use*).

Nalidixic Acid

Nalidixic acid together with high-dose intravenous melphalan has caused deaths in children due to haemorrhagic enterocolitis.

Busulfan

In paediatric population, for the Busulfan-Melphalan regimen it has been reported that the administration of melphalan less than 24 hours after the last oral busulfan administration may influence the development of toxicities.

Cyclosporin

Impaired renal function has been described in bone marrow transplant patients who were conditioned with high-dose intravenous melphalan and who subsequently received cyclosporin to prevent graft-versus-host disease.

Fertility, Pregnancy and Lactation

Fertility

Melphalan causes suppression of ovarian function in premenopausal women resulting in amenorrhoea in a significant number of patients.

There is evidence from some animal studies that melphalan can have an adverse effect on spermatogenesis. Therefore, it is possible that melphalan may cause temporary or permanent sterility in male patients.

Women of childbearing potential / Contraception in males and females

Female patients who use Melphalan should use effective and reliable contraceptive methods during treatment and for a period of six months, following the cessation of treatment.

Male patients should use effective and reliable contraceptive methods during treatment and for a period of three months following the cessation of treatment.

The final decision regarding the additional time period on contraception should be taken by the doctor and / or the patient (*see Special Warnings and Precautions for Use*).

Pregnancy

The teratogenic potential of melphalan has not been studied. In view of its mutagenic properties and structural similarity to known teratogenic compounds, it is possible that melphalan could cause congenital defects in the offspring of patients treated with the drug.

As with all cytotoxic chemotherapy, adequate contraceptive precautions should be practised when either partner is receiving melphalan.

The use of melphalan should be avoided whenever possible during pregnancy, particularly during the first trimester. In any individual case the potential hazard to the foetus must be balanced against the expected benefit to the mother.

Breastfeeding

Mothers receiving melphalan should not breast-feed.

Effects on Ability to Drive and Use Machines

No data.

Undesirable Effects

For this product there is no modern clinical documentation which can be used as support for determining the frequency of undesirable effects. Undesirable effects may vary in their incidence depending on the indication and dose received and also when given in combination with other therapeutic agents.

The following convention has been utilised for the classification of frequency: - Very common $\geq 1/10$, common $\geq 1/100$, $< 1/10$, uncommon $\geq 1/1000$ and $< 1/100$, rare $\geq 1/10,000$ and $< 1/1000$, very rare $< 1/10,000$, not known (cannot be estimated from the available data).

Tabulated list of adverse reactions

Body System	Frequency	Side Effects
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Neoplasms benign, malignant and unspecified (including cysts and polyps)	Not known	Secondary acute myeloid leukaemia and myelodysplastic syndrome (<i>see Special Warnings and Precautions For Use</i>)
Blood and Lymphatic System Disorders	Very common	bone marrow depression leading to leucopenia, thrombocytopenia and anaemia
	Rare	haemolytic anaemia
Immune System Disorders	Rare	hypersensitivity ¹ (<i>see Skin and Subcutaneous Tissue Disorders</i>)
Respiratory, Thoracic and Mediastinal Disorders	Rare	interstitial lung disease and pulmonary fibrosis (including fatal reports)
Gastrointestinal Disorders ²	Very common	nausea ² , vomiting and diarrhoea; stomatitis at high dose
	Rare	stomatitis at conventional dose
Hepatobiliary Disorders	Rare	liver disorders ranging from abnormal liver function tests to clinical manifestations such as hepatitis and jaundice
Skin and Subcutaneous Tissue Disorders	Very common	alopecia at high dose
	Common	alopecia at conventional dose
	Rare	rash maculo-papular and pruritus (<i>see Immune System Disorders</i>)
Renal and Urinary Disorders	Common	blood urea increased ³
	Not known	acute kidney injury
Reproductive system and breast disorders	Not known	azoospermia, amenorrhoea
Vascular disorders ⁴	Not known	deep vein thrombosis and pulmonary embolism
General Disorders and Administration Site Conditions	Very common	pyrexia

1. Allergic reactions to melphalan such as urticaria, oedema, skin rashes and anaphylactic shock have been reported uncommonly following initial or subsequent dosing, particularly after intravenous administration. Cardiac arrest has also been reported rarely in association with such events.
2. Gastrointestinal effects such as nausea and vomiting have been reported in up to 30% of patients receiving conventional oral doses of melphalan. The incidence of diarrhoea, vomiting and stomatitis becomes the dose-limiting toxicity in patients given high intravenous doses of melphalan in association with haemopoietic stem cell rescue. Cyclophosphamide pretreatment appears to reduce the severity of gastro-intestinal damage induced by high-dose melphalan and the literature should be consulted for details.
3. Temporary significant elevation of the blood urea has been seen in the early stages of melphalan therapy in myeloma patients with renal damage.
4. The clinically important adverse reactions associated with the use of melphalan in combination with thalidomide and prednisone or dexamethasone and to a lesser extent melphalan with lenalidomide and prednisone include: deep vein thrombosis and

pulmonary embolism (see *Posology and Method of Administration* and *Special Warnings and Precautions for Use*).

Overdose

Symptoms and signs

Gastro-intestinal effects, including nausea, vomiting and diarrhoea are the most likely early signs of acute oral overdosage.

The principal toxic effect is bone marrow suppression, leading to leucopenia, thrombocytopenia and anaemia.

Treatment

General supportive measures, together with appropriate blood and platelet transfusions, should be instituted if necessary, and consideration given to hospitalisation, cover with anti-infective agents, and the use of haematological growth factors.

There is no specific antidote. The blood picture should be closely monitored for at least 4 weeks following overdosage until there is evidence of recovery.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

Pharmacotherapeutic group: antineoplastic and immunomodulating agents, antineoplastic agents, alkylating agents, nitrogen mustard analogues

ATC Code: L01AA03

Mechanism of Action

Melphalan is a bifunctional alkylating agent. Formation of carbonium intermediates from each of the two bis-2-chloroethyl groups enables alkylation through covalent binding with the 7-nitrogen of guanine on DNA, cross-linking two DNA strands and thereby preventing cell replication.

Pharmacokinetic properties

Absorption

The absorption of oral melphalan is highly variable with respect to both the time to first appearance of the drug in plasma and peak plasma concentration.

In studies of the absolute bioavailability of melphalan the mean absolute bioavailability ranged from 56 to 85%.

Intravenous administration can be used to avoid variability in absorption associated with myeloablative treatment.

In a study of 18 patients administered melphalan 0.2 to 0.25 mg/kg bodyweight orally, a maximum plasma concentration (range 87 to 350 ng/ml) was reached within 0.5 to 2.0 hours.

The administration of melphalan tablets immediately after food delayed the time to achieving peak plasma concentrations and reduced the area under the plasma concentration-time curves by between 39 and 54%.

Distribution

Melphalan displays limited penetration of the blood-brain barrier. Several investigators have sampled cerebrospinal fluid and found no measurable drug. Low concentrations (~10% of that in plasma) were observed in a single high-dose study in children.

Elimination

In 13 patients given oral melphalan at 0.6 mg/kg bodyweight, the plasma mean terminal elimination half-life was 90 ± 57 min with 11% of the drug being recovered in the urine over 24 h.

In 18 patients administered melphalan 0.2 to 0.25 mg/kg bodyweight orally, the mean elimination half-life was 1.12 ± 0.15 h.

Special Patient Populations

- **Renal impairment**

Melphalan clearance may be decreased in renal impairment (*see Posology and Method of Administration - Renal impairment and Special Warnings and Precautions for Use- Renal impairment*).

- **Elderly**

No correlation has been shown between age and melphalan clearance or with melphalan terminal elimination half-life (*see Posology and Method of Administration*).

Pre-clinical Safety Data

Carcinogenesis, mutagenesis

Melphalan is mutagenic in animals.

Fertility studies

In mice, melphalan administered intraperitoneally at a dose of 7.5 mg/kg, showed reproductive effects attributable to cytotoxicity in specific male germ cell stages and induced dominant lethal mutations and heritable translocations in post-meiotic germ cells, particularly in mid to late stage spermatids.

A study was performed to measure the total reproductive capacity of melphalan in female mice. Females received a single intraperitoneal dose of 7.5 mg/kg melphalan and were then

housed with an untreated male for most of their reproductive life span (a minimum of 347 days post-treatment). A pronounced reduction in litter size occurred within the first post-treatment interval, followed by an almost complete recovery. Thereafter, a gradual decline in litter size occurred. This was simultaneous with a reduction in the proportion of productive females, a finding associated with an induced reduction in the number of small follicles (see *Fertility, Pregnancy and Lactation*).

PHARMACEUTICAL PARTICULARS

List of Excipients

Tablet Core:

Microcrystalline cellulose

Crospovidone

Colloidal anhydrous silica

Magnesium stearate

Tablet Film Coating:

Hypromellose

Titanium dioxide

Macrogol

Incompatibilities

None known.

Shelf Life

The expiry date is indicated on the packaging.

Special Precautions for Storage

Store at 2°C to 8°C.

Nature and Contents of Container

Melphalan are white to off-white film-coated, round, biconvex tablets engraved "GX EH3" on one side and "A" on the other, supplied in amber glass bottles with a child resistant closure containing 25 tablets.

Special Precautions for Disposal and Other Handling

Safe handling of melphalan tablets:

The handling of melphalan tablets should follow guidelines for the handling of cytotoxic drugs according to prevailing local recommendations and/or regulations.

Provided the outer coating of the tablet is intact, there is no risk in handling melphalan tablets.

Melphalan tablets should not be divided.

Disposal:

Melphalan tablets should be destroyed in accordance with relevant local regulatory requirements concerning the disposal of cytotoxic drugs.

Version number of package insert: CCDS22/01/1025

Date of revision of package insert: 15 Oct 2025

**MANUFACTURER**

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